u and v are orthogonal iff $u \cdot v = 0$.

6.2 Orthogonal Sets

– A set of vectors $\{u_1, u_2, \ldots, u_p\}$ in $\mathbb{R}^n$ is said to be an orthogonal set if each pair of distinct vectors from the set is orthogonal, that is, if ~~$\boldsymbol{u}_i \cdot \boldsymbol{u}_j = 0$~~ whenever $i \neq j$.

$u_i \cdot u_j = 0$

Ex: $u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$, $u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$, $u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$ Show that $\{u_1, u_2, u_3\}$ is an orthogonal set.

Sol

$u_1 \cdot u_2 = 3 \cdot -1 + 1 \cdot 2 + 1 \cdot 1 = 0$

$u_1 \cdot u_3 = 3 \cdot -1/2 + 1 \cdot -2 + 1 \cdot 7/2 = 0$

$u_2 \cdot u_3 = -1 \cdot -1/2 + 2 \cdot -2 + 1 \cdot 7/2 = 0$

Each pair of distinct vectors is orthogonal, and so $\{u_1, u_2, u_3\}$ is an orthogonal set.

Theorem:

If $S = \{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in $\mathbb{R}^n$, then S is linearly independent and hence is a basis for the subspace spanned by S.

Proof:

If $0 = c_1 u_1 + \cdots + c_p u_p$ for some scalars $c_1, \ldots, c_p$, then

$$0 = 0 \cdot u_1 = (c_1 u_1 + \cdots + c_p u_p) \cdot u_1$$

$$0 = (c_1 u_1) \cdot u_1 + (c_2 u_2) \cdot u_1 + \cdots + (c_p u_p) \cdot u_1$$

$$= c_1 (u_1 \cdot u_1) + c_2 (u_2 \cdot u_1) + \cdots + c_p (u_p \cdot u_1)$$

($u_2 \cdot u_1, \ldots, u_p \cdot u_1$ are 0)

$$0 = c_1 \cdot (u_1 \cdot u_1)$$

Since u_1 is nonzero $u_1 \cdot u_1$ is not zero and therefore $c_1 = 0$. Similarly $c_2, \ldots, c_p$ must be zero. Thus S is linearly independent.

Definition:

An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.

Theorem:

Let $\{\mathbf{u}_1, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $\mathbf{y}$ in W, the weights in the linear combination

$$\mathbf{y} = c_1\mathbf{u}_1 + \cdots + c_p\mathbf{u}_p$$

are given by

$$c_j = \frac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \qquad (j = 1, \ldots, p)$$

Proof:

Sol'n

$y\cdot u_1 = 11$ $\quad y\cdot u_2 = -12$ $\quad y\cdot u_3 = -33$

$u_1\cdot u_1 = 11$ $\quad u_2\cdot u_2 = 6$ $\quad u_3\cdot u_3 = 33/2$

By the above theorem:

$$y = \frac{y\cdot u_1}{u_1\cdot u_1}u_1 + \frac{y\cdot u_2}{u_2\cdot u_2}u_2 + \frac{y\cdot u_3}{u_3\cdot u_3}u_3 = \frac{11}{11}u_1 + \frac{-12}{6}u_2 + \frac{-33}{33/2}u_3$$

$$y = u_1 - 2u_2 - 2u_3$$

Ex: $S = \left\{u_1 = \begin{bmatrix}3\\1\\1\end{bmatrix}, u_2 = \begin{bmatrix}-1\\2\\1\end{bmatrix}, u_3 = \begin{bmatrix}-1/2\\-2\\3/2\end{bmatrix}\right\}$ Express $y = \begin{bmatrix}6\\1\\-8\end{bmatrix}$ as a linear combination of the vectors in S.